GFI – Ford Electrical Connections

Taps and Intercepts

There are two ways of making the electrical connections that are required by the GFI system.

A TAP is made by splicing a wire to a signal in the vehicle allowing the original connections to remain in place. With a TAP, the signal stays connected to the vehicle in all modes of operation. The GFI system can read the signal (e.g. a sensor) and modify it if necessary (e.g. a bi-directional data bus). A TAP is usually implemented by pulling a wire out of a connector and joining it to a GFI harness. A wire from the GFI harness is then inserted into the connector. The harnesses are designed to provide matching wire colors for TAPped connections.

An INTERCEPT is made by cutting a signal path so that the source and load are routed independently to the GFI system. An INTERCEPT is made when the GFI system must completely disable a device (e.g. fuel pump) or drive a device independently of the vehicle's normal method (e.g. fuel gauge) during alternate fuel mode. During other modes, the original connections are restored. An INTERCEPT is usually implemented by disconnecting a pair of connectors in the vehicle's harness system and joining them to a matching pair of connectors included in the GFI harness. The signals of interest are routed to the GFI system. Any remaining signals are routed directly between the new connectors.

Taps

Tach or Crank Position Sensor (2 wires)

MIL

Bus + and Bus –

Left Front HEGO, Right Front HEGO

Throttle Position Sensor

Signal Return

Switched Battery (Hot start and run)

Battery

Intercepts

Fuel Pump Power

Fuel Gauge

Instrument Cluster Power

Notes about Fuel Gauges

These are handled in two ways.

1999 and later F-Series vehicles have their instrument cluster connected to the SCP bus (Bus + and Bus –). These vehicles do not require intercepts in the fuel gauge and instrument cluster power signals because the GFI system can send fuel level data over the SCP bus.

All other vehicles (earlier F-Series, P131, Econoline vans, Contour) require fuel gauge and cluster power intercepts to allow GFI to control the fuel gauge in alternate fuel mode and reset the cluster at fuel switchover.